

Appendix A: Speed of Light c

In SAT, c emerges as the maximal causal speed from the foliation structure and strain tensor, without external input.

Appendix B: Planck's Constant $\hbar$

In SAT, $\hbar$ emerges as the quantum of action from the compactness of ψ , without external input.

Appendix C: Elementary Charge e

In SAT, e emerges as a quantized coupling constant from the ψ - $U(1)$ structure, without external input.

Appendix D: Fine-Structure Constant α

In SAT, α emerges necessarily as a dimensionless ratio of internally derived quantities e , $\hbar$, and c , without external input or tuning.

Appendix E: Newton's Gravitational Constant G

In SAT, G emerges from the dynamics of the foliation strain tensor $S_{\mu\nu}$, without external input or tuning.

Appendix F: Electron Mass m_e

In SAT, the electron mass m_e emerges as the minimal excitation of the internal clock phase winding, determined by the fundamental clock frequency ν_0 , without external input or tuning.

Appendix G: Bohr Radius a_0 and Rydberg Constant R_∞

In SAT, a_0 and R_∞ emerge as combinations of e , $\hbar$, m_e , and c , without external input or tuning.

Appendix H: Hydrogen Molecule Dissociation Energy $D_0(H_2)$

In SAT, $D_0(H_2)$ emerges as a fraction of the Hartree energy, constructed from e , $\hbar$, m_e , and c , without external input or tuning.

Appendix I: Standard Model Gauge Group $SU(3) \times SU(2) \times U(1)$

In SAT, the Standard Model gauge group emerges naturally from the topology of $\psi(x)$ and $\tau(x)$, and the geometry of $u^\mu(x)$, without external input.

Appendix J: Charge Quantization

In SAT, electric charge quantization follows from the compactness of $\psi(x)$ and invariance under large $U(1)$ gauge transformations, without external input or tuning.

Appendix K: Anomaly Cancellation

In SAT, anomaly cancellation follows from the topological and combinatorial structure of matter fields under $SU(3) \times SU(2) \times U(1)$, without external input.

Appendix L: Yukawa Structures and Mass Hierarchies

In SAT, fermion mass hierarchies emerge naturally from the combinatorial winding structure of $\psi(x)$ and the topological sectors of $\tau(x)$, replicating Yukawa hierarchies without external input.

Appendix M: Falsifiable Predictions (Clock Drift, Domain Walls, Pulsar Timing)

In SAT, measurable effects such as clock drift, domain wall phase shifts, and pulsar timing residuals emerge from the internal dynamics of $\theta_4(x)$, $u^\mu(x)$, and $\psi(x)$, offering direct, falsifiable experimental tests.

Appendix N: Neutrino Mass Suppression

In SAT, neutrino masses are naturally suppressed relative to charged fermions due to the absence of topological enhancement from $\tau(x)$, leading to small but nonzero masses without external input.

Appendix O: Proton Stability

In SAT, proton stability emerges from the topological conservation laws associated with $\tau(x)$, forbidding baryon-number-violating processes without external symmetries or tuning.

Appendix P: Gravitational Wave Modifications (LIGO Scale)

In SAT, high-frequency gravitational wave propagation is modified due to strain-induced nonlinear dispersion from $u^\mu(x)$, potentially observable at LIGO frequencies, without external input or tuning.